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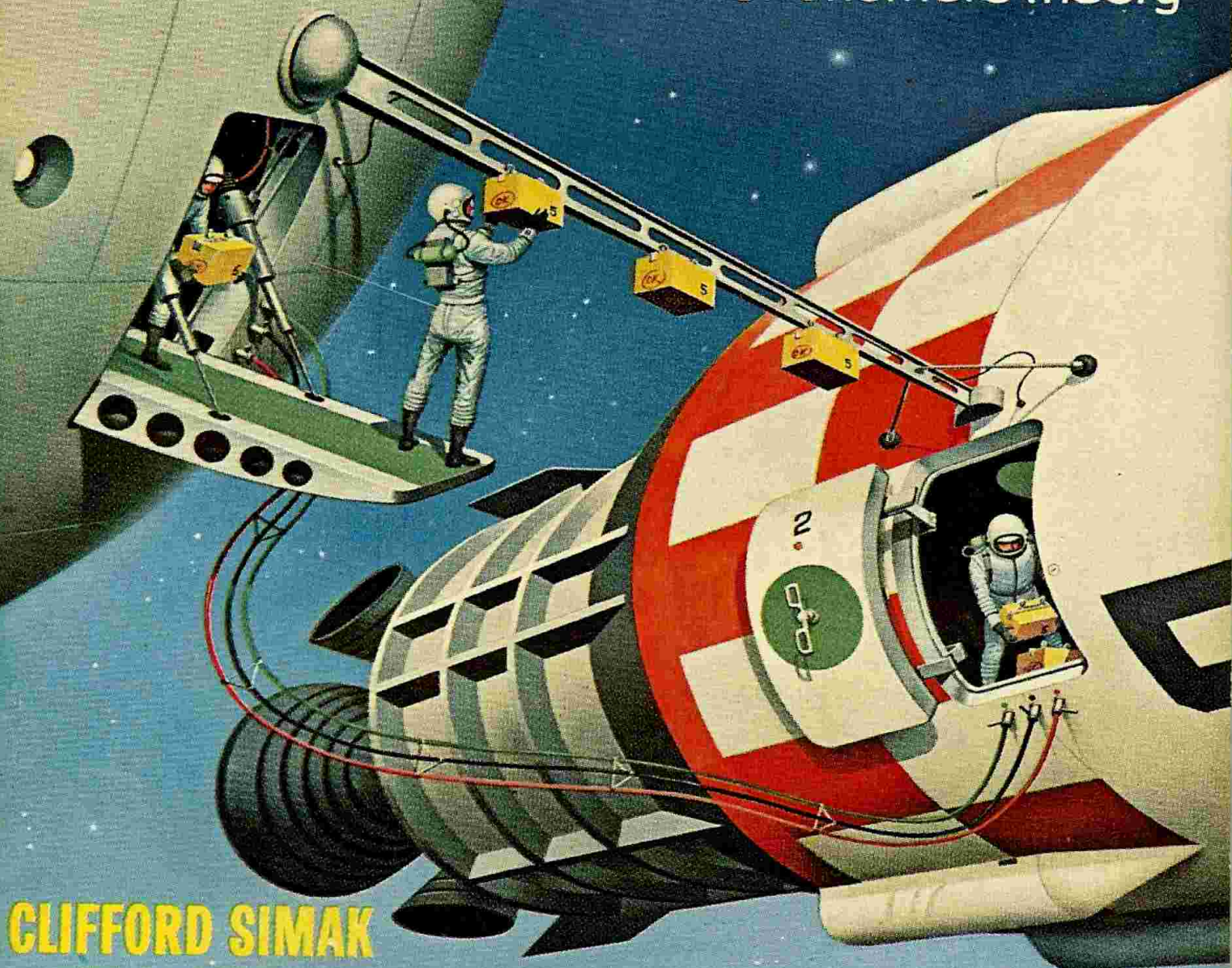
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THUNDER IN SPACE by **Lester del Rey**

Extra-Terrestrial Life: An Astronomer's Theory



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--SF PROFILE
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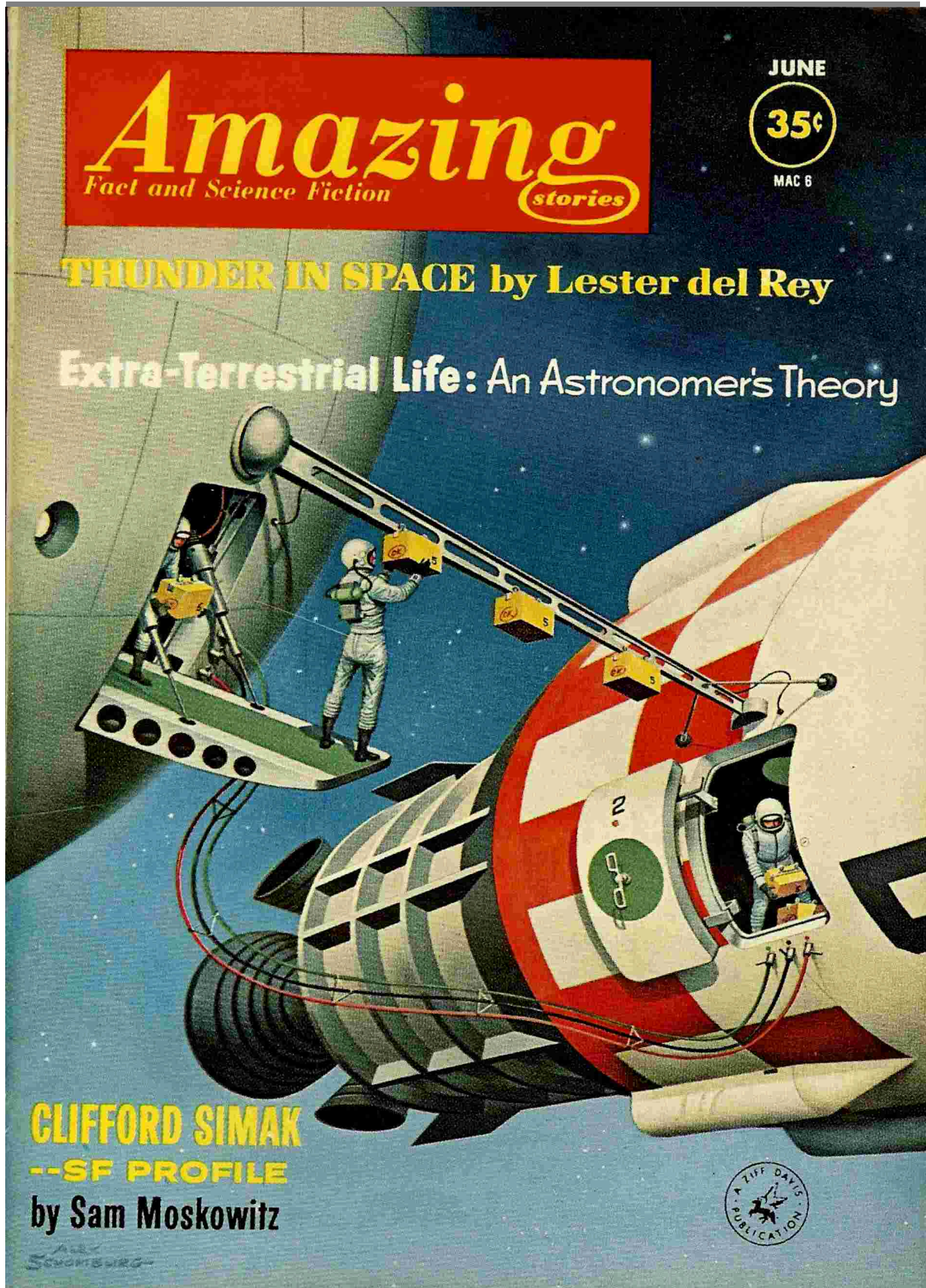
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*** START OF THE PROJECT GUTENBERG EBOOK THUNDER IN
SPACE ***

The men on the space station had a word for trouble—"thunder." Always it had been thunder on earth. Now, with the warheads decaying and the Soviets playing a mysterious game, now there was ...

THUNDER in SPACE

By LESTER del REY

Illustrated by FINLAY

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I

In the little formal garden in Geneva, the guards had withdrawn discreetly, out of sight and hearing of the two men who sat on a carved marble bench in the center of the enclosure.

The President of the United States was too old for the days of strained public and private meetings and the constant badgering of his advisers that had preceded this final, seemingly foredoomed effort. His hands trembled as he lifted them to light a cigarette. Only his voice still held its accustomed calm.

"Then it's stalemate, Feodor Stepanovich. I can make no more concessions without risking impeachment."

The dark, massive head of the Russian Premier nodded. "Nor can I, without committing political suicide." His English was better than the rural dialect of Russian he still retained. "Call it a double checkmate. Our predecessors sowed their seeds too deep for our spades. Or should I say, too high?"

Both heads turned to the north, where a bright spot was climbing above the horizon. The space station sparkled in sunlight far above Earth, sliding with Olympian deliberation past a few visible stars until it was directly overhead. Without a timetable or a telescope, there was no way of knowing whether it was the Russian *Tsiolkovsky* or the American *Goddard*, nor did either man care. Half the world lived in almost hysterical fear of one or the other, with the rest of the human race existing in terror of both.

The Premier muttered something from the ugliness of his childhood experiences, but the President only sighed unhappily, as if sorry that his own background gave him no such expressions.

A few minutes later, the leaders separated. As they moved across the garden, their escorts surrounded them, clearing the way toward the cars that would take them to the airport. Behind them, professional diplomats stopped puzzling over the delay and began spinning obfuscations to cynical reporters. The phrases had long since lost all meaning, but the traditions of propaganda had to be maintained.

In the UN, the Israeli delegate crumpled a news dispatch and began speaking without notes, demanding that space be inter-nationalized. It was the greatest

speech of his career, and even the delegate from Egypt applauded. But national survival could not be trusted to the shaky impartiality of the UN. The resolution was vetoed by both the United States and Russia.

The Fourteenth Space Disarmament Conference was ended.

II

A month later, a thousand miles above Earth and exactly 180° behind the *Tsiolkovsky*, the *Goddard* swung steadily around the globe in a two-hour circumpolar orbit. Outwardly, it looked like the great metal doughnut that space artists had pictured for decades. On the inside, however, the evidence of hasty, crash-planned work was everywhere. The air fans whined and vibrated, the halls creaked and groaned, and the water needed to maintain balance gurgled and banged through ill-conceived piping. It was cramped and totally inadequate for the needs of the nation that had put it into space eight years before in a rush attempt to match the Russian "Sulky".

Jerry Blane should have been used to such conditions. He'd been one of the original space-struck men who'd helped to build it and then had been lucky enough to get a permanent assignment. Now he drifted in the weightless hub, watching the loading of a ship bound back for the home planet, wondering what hell's brew the boxes contained. The project that had usurped the cryogenic labs had involved its own crew of scientists, who were already on board the ship, taking their secret with them.

He shrugged, trying to dismiss the problem. The motion twitched him about, and he corrected automatically. His tall, thin body was accustomed to weightlessness.

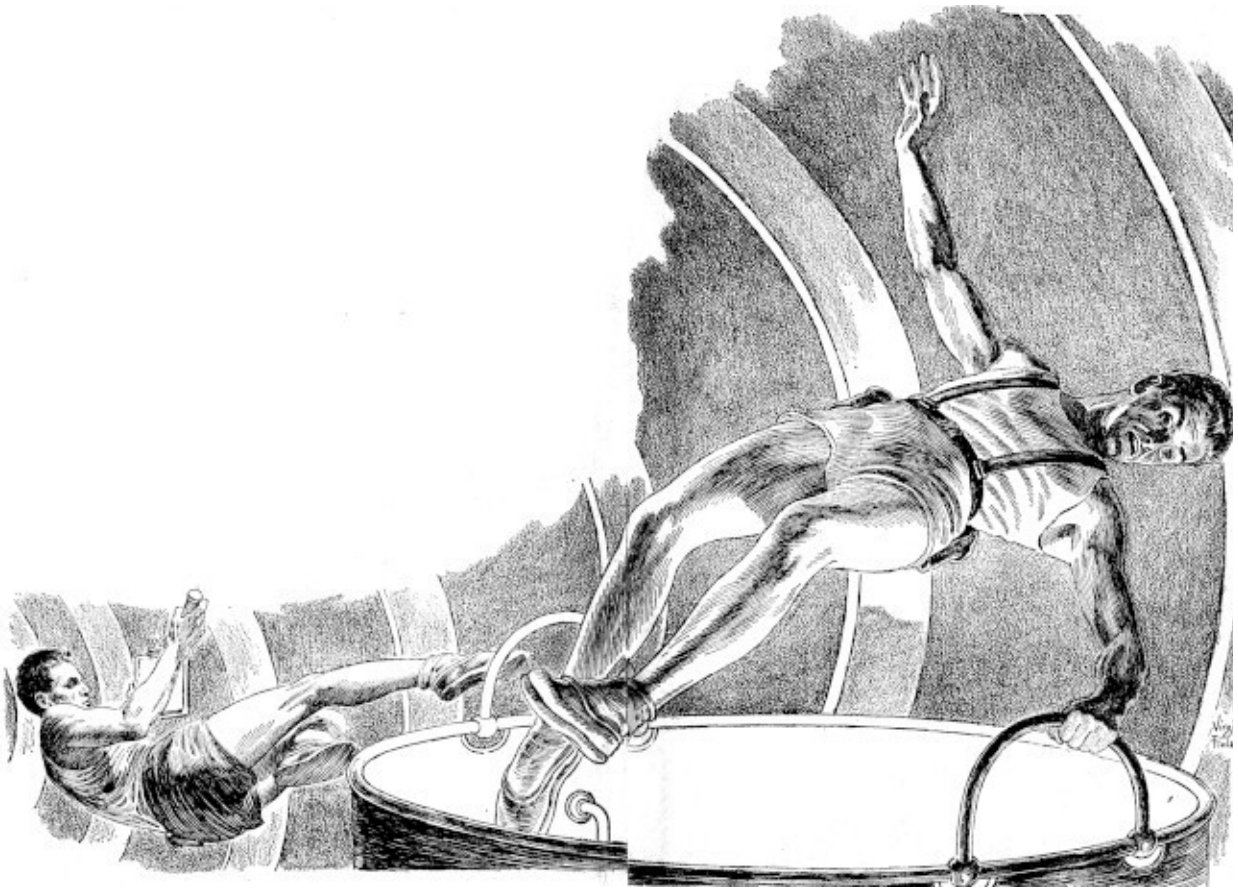
Beside him, the head of the science corps on the station also floated in midair. The big body of Dr. Austin Peal was revealed in the single pair of shorts customary on the *Goddard*, and its darkness contrasted sharply with the blond hair and pale skin of Blane. Only the frowns matched.

The short, intense figure of General Devlin popped into the hub from the tube elevator ahead of the pilot, Edwards. In spite of the weightlessness, the

station commandant managed to pull himself to rigid attention at sight of Blane. He scowled, but held out his hand with formal correctness.

"All right, Blane. You're in charge officially until I get back," he admitted grudgingly. He obviously resented the order that left a civilian in charge while he went down to testify for the station appropriations and receive new orders. "You'll find detailed notes on my desk. I suggest you follow them to the letter."

He grabbed a handhold and began pulling himself into the airlock to the ship without waiting for a reply.



Edwards had lingered. Now he also held out his hand. "Wish me luck, Jerry," he said. "I may need it."

Because of the contents of the boxes and the presence of Devlin, Edwards had been ordered to make his landing at Canaveral, under military security. Most space work was done from Johnston Island in the Pacific; the inadequate facilities at the Cape were supposed to be used only by smaller rockets. But lately the rules were shot in a lot of ways. Ever since the last meeting at Geneva, nothing seemed normal.

"You'll make out," Drake told him. "Our predictions give you perfect landing weather, at least."

"Yeah. Clear weather and thunder below." In the station slang, thunder stood for heavy trouble. The weather forecast didn't matter; there was always thunder below.

Edwards moved through the airlock and into his ship. A moment later, fire bloomed from the rocket tubes and the ship began moving away. In the station, motors began whining, restoring the hub's spin to match that of the rest of the *Goddard*.

From the viewing ports, Earth filled almost the entire field of vision, like a giant opal set in black velvet. More than half was covered by bright cloud masses, but the rest showed swirls and patterns of blue water, green forest and reddish brown barren patches. Over everything lay the almost fluorescent blue of atmosphere, forming a brilliant violet halo at the horizon. It looked incredibly beautiful. So, Blane thought, does a Portuguese man-of-war—until one sees the slime underneath or touches the poisoned stings.

"Why can't they leave us alone?" Peal asked, as if reading Blane's mind. "Why can't they blow themselves up quietly without ruining our chances here?"

Blane chuckled bitterly. He'd been on vacation down there a month before, and Earth was fresher in his memory than it was to Peal. "They don't see it that way. To them, we're the danger, the biggest sword of Damocles ever invented. They look up and see us going overhead, loaded with enough

megaton bombs to blast life off Earth. Every time we orbit over them, they see Armageddon right over their heads, waiting some fool's itching finger. They could risk the holocaust when everything was halfway around the world, but not when it's where they can look up and see it. Most of the thunder down there is caused by the chained lightning we're carrying up here."

It wasn't an original idea. The panic on Earth had been increasing since the building of the Russian station. Now panic bred false moves, and errors bred more panic. Sooner or later, that panic could get out of hand and bring about the very ruin they feared.

"Besides," he added, "there's the expense of keeping us up here. They think the billions needed to maintain us are pauperizing them."

"We're paying three to one on every cent we get! Even forgetting the work in astronomy, bio-chemistry, cryogenics and high-vacuum research, our weather predictions are worth billions a year in crop returns."

Blane shrugged. "Most of our work is for the government without payment, so Congress still has to appropriate billions for us yearly. That's all the people see. We're poison down there. They'd vote to ditch us if they weren't so scared of the bombs on the *Sulky*."

"That's what comes of putting scientific tools under government control," Peel grumbled. "The stations should have been private enterprises from the beginning."

Blane nodded automatically. It was an old argument, and it made sense. But there was no chance of the government ever letting go now. They took the clanking elevator down toward the rim, while weight built up to the normal one-third Earth gravity that was produced by the spin at the outer edge of the *Goddard*. Then they moved along the hallway that circled the rim, through the recreation hall, past the vacuum labs that were busy with some kind of military development, and past the cryogenic section, where men were busy getting ready to resume normal work. Beyond that lay the weather study section. It should have been located in the hub, but there had been too little room, and the pickups were remotely controlled, flashing their pictures of

Earth onto big screens here. Now the screens showed Madagascar to the west of them as they swung northward. Men were busy plotting the final details for next month's weather predictions.

Peal followed Blane through the side door into the little office of Devlin. The General was something of a martinet, but his discipline extended to himself. Everything was in order, and the list of instructions lay in a folder in the center of the desk. Blane glanced at it, then at the basket of communications from Earth. He grimaced, and passed some of the flimsies over to Peal. "There's more evidence, if you want to prove the profit we could show."

There were requests for projects to be done here, complaints—often angry—at projects already okayed but delayed by high-priority military research. There were applications from names already famous below. Five foundations were demanding that the lunar ships be rushed to completion.

The intercom came to life with a rasping parody of the voice of Devlin's secretary. "Mr. Blane, Captain Manners insists on seeing you. He's been waiting nearly an hour."

Blane flipped through Devlin's instructions. There was an entry on Manners there: *Troublemaker, possibly paranoid. Add his figures to HQ report as routine only.*

"Send him in," Blane ordered. The red-headed young captain had been assigned here only six months ago, but Blane had met him often enough to like him.

Almost at once, the connecting office door opened and Manners shoved in. He was obviously angry, but his voice didn't show it. "Thanks for seeing me, Blane. I'd just about decided you wouldn't." He slapped a piece of film down on the desk. "Here. Look at that!"

The film was slightly darkened. Blane turned it over, recognizing it as one of the strips worn by the men who worked in the bomb section to warn of any accidental exposure to radiation. But it was well under any dangerous level of exposure. He passed it to Peal, who studied it in curiosity.

"That's in five hours of routine work in the bomb bay," Manners said. "Routine work! And I checked the films before issuing them, so I know they weren't pre-exposed." He pulled out a sheet of paper covered with figures

and dropped it on the desk. "The radiation's up in there again. Check it yourself if you won't accept my readings."

Peal had grabbed up the figures which listed the radiation count in various sections of the bomb bay. They meant nothing to Blane, but the scientist tensed visibly as he studied them.

"I gather you showed your figures to Devlin," Blane said. "What did he say about them?"

Bitterness washed over Manners' face. "He told me to forget it, that readings were higher here than what I'd learned handling warheads below because we got so many cosmic rays. Three months ago, they were a lot higher, and he said there was an increase in cosmic radiation. But he okayed my getting the air pumped out of the bay so nothing hot would be sucked into the rest of the station. Last month, the figures went up to about half what they are now, and he mumbled something about a cosmic ray storm. I haven't been able to see him since then."

"There's no such thing as a cosmic ray storm," Peal said flatly. "Why wasn't this reported to me? It's partly in my province."

"General Devlin ordered me not to discuss it with anyone!"

"Thunder?" Blane asked the scientist.

"If it keeps doubling every month, it's disaster! The thin walls here are no protection from radiation. Even now, we'd better evacuate the bio labs beside the bay. Captain Manners, we'll have to check you on this. I'm not exactly doubting your word, but these results are impossible according to anything I know." He swung to Blane. "I think you'd better come, too, Jerry. This may be something for the authorities, and you carry the weight here now."

It was a lousy beginning to his temporary command, Blane thought. But seeing Peal's face, he simply nodded and followed the other two out into the hall. They were heading toward the bomb section when a shout went up from some of the men watching the viewing screens.

Blane swore to himself, but turned back.

He saw at once that the screens were set for top magnification, showing a section of Earth at the extreme limits of resolution. A glance at the projected coördinates showed that they were over southern Russia. His eyes were untrained at grasping details, but he saw enough to recognize that they must be viewing the great Russian rocket base that supplied the *Sulky*.

Scarfield had taken over from his subordinate and began picking out details with a moving spot of light. "Rocket—see its shadow? And there—there—there. Jerry, they've got every ship they own assembled together. And it looks as if they've been running supplies to them all. Something big's due."

"Attack?" Blane asked. One of the jobs of the station was to spot any clustering of military rockets that might presage a ground-based attack.

Scarfield shook his head. "Not a chance. Those are space rockets, not war missiles. This is like the massed flight they sent up about two years ago, remember? We never did figure out why they had to take the whole fleet out. But with what's going on below, this must mean something important. Think we should alert HQ?"

They obviously should, as soon as they were over one of their own stations. The rule was clear on that—when in doubt, shout! But meantime, they'd have to watch while still in view.

There was a faint spot of light, and Scarfield grunted. "They're blasting off! Maybe we can plot orbits and—"

The bright spot split into lances of fire, exploding savagely outwards! Every drop of monopropellant in the tanks must have let go at once to make such a flare. Then, before Blane could catch his breath, there was another flare and another. Suddenly the whole field was a great spread of flame as the other rockets were exploded by the savage blast of the first.

Before the *Goddard* had passed beyond view, they knew that every Russian ship on the field was totally demolished—which meant, according to Scarfield's estimate, every ship that could make the trip up to the *Sulky*.

They stared at the screen in shocked silence while Blane slowly began to realize the implications. It had happened while they were directly overhead. What would that mean to the ever-suspicious people of Russia who were already conditioned to think of the *Goddard* as their greatest enemy? What could be made of that in a world already close to the edge of panic?

III

By the time the *Goddard* was over the North Pole where she could make radio contact with Alaska, the news was already out. For once, Tass had released the news of a catastrophe without delay. The ground radio confirmed the fact that every supply ship for the *Sulky* had been wiped out, and that the detonation had been so great that even ships being assembled nearby had been wrecked hopelessly. It would be three months before Russia could again reach her station.

Later news filtered in slowly. Most of it had to be picked up from the regular FM news broadcasts that filtered through the ionosphere. A couple of the scientists who had learned Russian interpreted the news from Radio Moscow on their next trip over.

Surprisingly, there were no claims of American sabotage. Then Blane wondered whether it was so surprising. With the level of fear in Russia as high as elsewhere, it would probably have been a grave mistake for the leaders to suggest that any American sabotage of territory so far inside Russia was possible. The people had to count on the invulnerability of their station for what little hope they had; how that worked when the supply ships were already ruined was more than he could guess, but he had long since given up trying to understand the devious game of propaganda being played on Earth.

At least for the moment, the disaster was not being turned into another excuse to push the seemingly inevitable war another millimeter closer to the brink. Maybe the whole affair might result in some decline of tension. Once the American ships were sent up to supply the *Sulky* on an emergency basis, there might be a little good will from Russia and self-satisfaction at a good deed in America. That could give a breathing spell.

Blane had almost forgotten Manners and the worry over the strange increase in radioactivity. He had sent Manners' latest figures down with a query for instructions at the first chance to do so by tight-beam radio that would not leak security, and then had let the matter drop from his mind. It was several hours later when his secretary announced that Peal and Manners were in the outer office.

Manners looked both more worried and strangely satisfied, as if he were bursting to cry his I-told-you-so. But Peal's face was drained of any emotion except surprise.

The scientist nodded. "Captain Manners' figures were quite accurate. We've got to evacuate nearby sections of the station. In a way, we're lucky—radiation travels in straight lines, and the hull curves away from it here. There is about three hundred times normal radiation in there, and it's coming from inside the warheads. It isn't lethal yet—men can work there for a few hours at a time; but at the rate it's increasing, it soon will be. Any word from Earth?"

Blane dug through his in basket, and finally located a blue slip. It was in code, but Devlin's instructions included the location of the code book. He riffled through it for phrases each decagraph covered. *Situation within normal expectations—results being studied here—continue as at present—will apprise if new procedure advisable—regard as utmost top secret—invoke maximum security measures over affected personnel.*

"No word," he said bitterly. Probably he wasn't even supposed to say that much, or to discuss it with the other two. But he chose to interpret the part about continuing as at present to permit the discussion to continue. He tried to focus his mind on what facts he knew. "I thought the radiation rate of the stuff in the warheads was constant, and that the casings were adequate shielding."

Peal nodded. "That's what's driving me out of my mind at the moment, Jerry. Except when it reaches critical mass, uranium-235 is supposed to have an absolutely fixed half-life; it shouldn't increase under any circumstances, and the mass of each section in those bombs can't increase to become nearer critical, either. It simply can't happen, according to any physics I ever learned. But it's doing so."

"What about the effects of cosmic rays?" Blane asked. Devlin might have learned more from Earth, and even if his story to Manners had been patently untrue, it might still offer some clue.

Peal shook his head, but somewhat doubtfully. "On Earth, they're mostly only mesons from strikes by cosmic radiation. Out here, we get only the extremely hard radiation—the shielding of the ship is too thin to affect them. Maybe they might speed up the half-life a little—but they shouldn't make it increase. I've been thinking about them, too. Meteorites show a much greater decay of uranium to lead than the ores on Earth, which might indicate some effect from cosmic radiation. But unless they somehow produce another isotope from uranium that's raising the activity, I can't figure it out. We need a top level nuclear physicist for this, and we don't have one here."

They discussed it at greater length, but without adding anything to their speculations. Blane felt the hairs on the back of his neck prickling, and was conscious of a vague picture in his mind of the warheads ticking away and getting set to blast spontaneously. But he put the idea aside. Earth might be a little careless of their welfare under the pressure of emergency, but right now Earth would never risk losing the station. It was only his overactive imagination.

He finally assigned Peal and Manners back to the task of studying the matter as best they could, and tried to dismiss it from his mind. There were more than enough other worries about the station. The cryogenics lab was in trouble—the group from Earth who had used the labs had badly depleted supplies and been careless about equipment that was common enough below but difficult to obtain here. The evacuation of the laboratories near the bomb bay threw severe strains on research, and Earth was demanding that some of it be speeded up. And the weather study was being crippled by the need to waste too much attention on detailed studies of every section of Russia. The whole station was on emergency orders to do twice as much as could possibly be done.

He waited for news that supplies were being sent from Johnston Island to the *Sulky*, but no such news appeared. Instead, the news carried details that were only rumors of some effort of the United States to force Russia to disarm the *Sulky* unilaterally in return for the loan of eight rocket ships and launching facilities. If such an offer had been made, it must have been turned down flatly. The next day there was not even a mention of it.

When Edwards came up again, Blane sent for him at once. The pilot had made a superb landing of his ship at Canaveral, and had then been jetted back to the Island. Normally he would have taken a long layover there before making another trip up, though he had senior pilot's right to select or refuse any flight he chose. Blane was curious about his reasons for choosing the first trip he could make.

Edwards lost no time in reporting. He hadn't stopped to remove his emergency space suit, though he'd left the helmet and the oxygen tank somewhere. He clumped in, accepted coffee, and began talking even as he shucked off the suit.

"It's a wonder they even let me fly up supplies to you," he grumbled. "Jerry, it's rough down there. They've got everything sewed up under controls. I'm surprised they didn't suspect me of plotting an orbit for the *Sulky* instead of here. Damn all governments that have to mess into space affairs!"

Some of the details came out slowly, with more color than clarity. But Blane gathered that they had reacted violently to the news that the government was trying to use the emergency as a means of forcing disarmament on the Russian station.

"You mean they actually did refuse help without such an agreement?" Blane asked. He hadn't wanted to believe the rumors.

Edwards nodded angrily. "They issued a ban against any efforts to help without such agreement. They most certainly did! And you can guess how that sat with us. Maybe the *Sulky's* full of Russians, but they're Russian *spacemen*! Hell, when we were building this wheel here and one worker got thrown out into space, three of their pilots came up in ships to help find him—and one did find him. Remember? Sure you do. They hated our building here, but they wouldn't let a man die in space if they could help. So we owe them a few trips."

Two of the pilots had tried to steal one of the ships fueled and supplied for the *Goddard*, but had been caught before they could take off. Now they were under guard, and the ships were being watched carefully. Edwards had been permitted to make the run only after a session in which it was pointed out

that landing rights would be denied any ship contacting the *Sulky*. And the other pilots were almost in a state of revolt, with nearly all of the old-time ground force supporting them.

"The government can't stick to such a policy," Blane said doubtfully. "They can't gain anything. The *Sulky* must have enough supplies for existence until at least one ship can be assembled and sent up. All we'll do by holding them up is to increase the danger. They must be bluffing for a while, hoping Russia will crack, but ready to send supplies in a few days."

Edwards stared at him in surprise. "You mean you don't know?" Then he slapped his thigh in disgust. "No, of course you don't. I keep forgetting you couldn't. The *Sulky* couldn't reach you by radio with the Earth in between. Jerry, we got a beamed message on the Island from her when she went over one time. SOS. She's in trouble right now. Can't get help from her base, and can't wait for negotiations, so she tried calling us direct. Security clamped down on the message at once, but the radio operator's as much space as we are, so he made a dupe copy for the pilots. The day after the blowup at the base, the *Sulky* ran into a meteoroid big enough to rip out part of her solar boiler. She lost most of the mercury into space, and the rest isn't enough, even when she's patched. She has to operate on batteries right now, and that won't last more than another day or so."

Blane winced at the picture. A station was dependent on power for its existence. Lights, air circulation, water for balance, heat regulation, and even the growing of plants to keep the air breathable depended upon a steady supply of power. Like the *Goddard*, the *Tsiolkovsky* used a reflecting trough on top that directed the intense solar radiation onto a pipe filled with mercury which was heated to gaseous form and operated the boiler and generator. It was far cheaper and safer than atomic power.

"The government knew of that when it refused help?" he asked incredulously.

Edwards grunted. "Didn't start their extortion plans until they knew!" Then he grinned slowly. "Funny thing, Jerry, when I checked over the supplies I brought up for you, I found some of the boxes of equipment got mixed up in shipment. They're full of cans of mercury! I left them aboard the ship, figuring you wouldn't need them here."

Blane found his face muscles were trying to frown and smile at the same time, and he caught himself before he could laugh. He went to the door to make sure it was locked, and came back to his desk slowly.

In theory, it was entirely possible to reach the *Sulky* from the *Goddard*, and every pilot knew the general orbit. The *Sulky* and the *Goddard* each took two hours to circle Earth, with one an hour behind the other. If a ship took off outward with a reasonable use of power it could get into an ellipse around Earth that would take three hours to bring it back to its starting point—and by then, the opposite station would be at that point. The maneuver could be made both ways with the fuel a final stage could carry easily enough.

"You don't have fuel enough," he decided.

"Nope. But you do—out in the blasted lunar ships that are still waiting appropriations."

Blane hadn't had time to think of the lunar ships during the hectic days of commanding the station. But Edwards' statement was true enough. The ships had been nearing completion for the long-desired American exploration of the Moon a year ago when Congress had eliminated appropriations for everything not connected with the current emergency. They still trailed the station a few miles in space. The workers had all returned to Earth, but the fuel still lay in the plastic balloons. The little ferry ship used between the ships and the station was still here, too. It could be used to bring the fuel back easily, since it had been equipped with tanks for moving fuel between supply rockets and the balloons.

"It wouldn't work," he said at last. "They'd spot your ship from Earth if you took off for the *Sulky*. They'd even guess where you'd gone when you didn't return on schedule. They might even refuse to let you land, and they'd probably make things impossible up here, too."

"I'll take my chances—and so will you," Edwards protested.

"Not unless it's necessary. Sure, somebody's got to make the trip. But it doesn't have to be your ship. The ferry's a lot smaller, but it can handle that much cargo and fuel on such an orbit." He grinned at Edwards' stubborn expression. "Look, you know I ran it for a year while we built the station. I can still pilot it, and Austin Peal can handle the math in computing the orbit.

I'll get it over to you and you can transship the mercury, then take off on schedule. Then let Earth guess what happens."

"And what will they do to you if they find out?"

"Nothing—officially. Nobody has told me officially that the policy is against offering help, so I'll proceed in terms of the older tradition. When you let slip the trouble on the *Sulky* and I found cans of mercury stored in the ferry, what could I do but assume the station was expected to get them to the other station?" Blane grinned, feeling sudden relief from his other worries. "Besides, I don't give a darn what they do to me. I'm only temporary boss here."

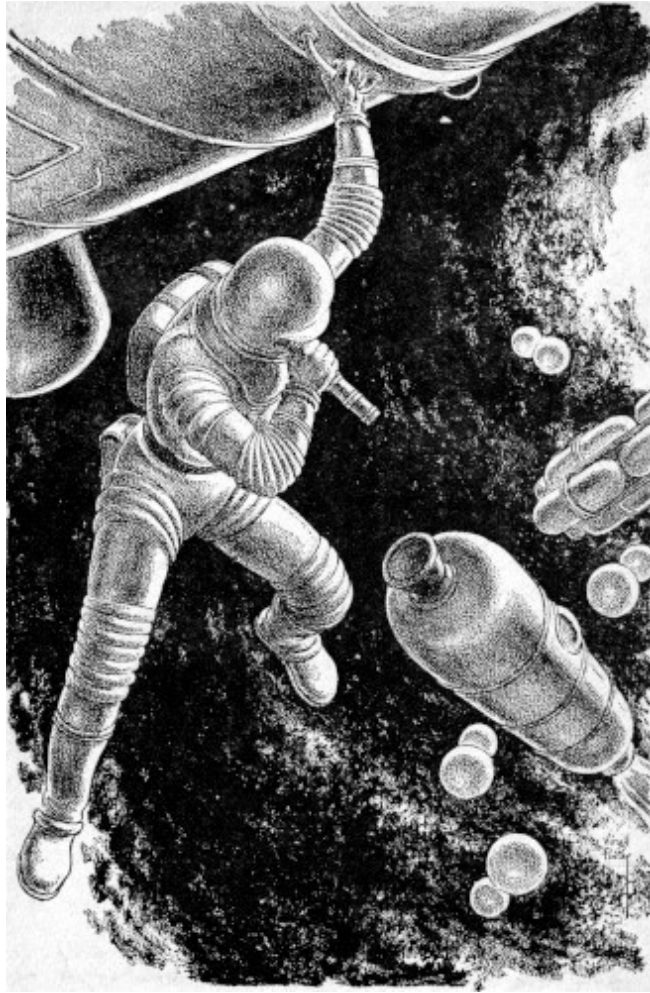
Edwards nodded. "I'll take your last reason, Jerry. Only don't bother moving the ferry. I can work it over beside my ship, and it'll make your explanation sound better. Good luck. And if you do get in a jam—all the guys will be on your side."

He went out while Blane started off to find Peal. He had doubts about involving the scientist now. The man had never been part of a real space team. Yet someone had to do the preliminary computing. He had more doubts as he tried to explain things to Peal; the man listened quietly, making no comment, and with no visible approval or disapproval.

When Blane finished, Peal stood up, nodding. "Thanks for letting me in on it, Jerry. You get the fuel and I'll have the computations off the calculator by the time you get back here."

IV

The ferry was a sausage-shaped structure of thin metal and plastic with an airlock at the front and a small reaction motor at the rear. It had been modified to hold either solid or liquid cargo and to operate off the monopropellant fuel instead of the lox and kerosene used when the station was built. There was even a plastic pipe between the cargo tank and its fuel tank to save separate filling, and no further modification was needed.



Blane took it out after checking the stowage of the mercury cans. He was slightly rusty, but he steadied down as he jockeyed into position beside one of the three lunar ships. He'd picked a balloon on the sunward side, and the warm fuel was soon flowing into his tank, forced through a long tube by a tiny, built-in pump. When he took off again, the ferry was overloaded and sluggish, but it showed no evidence of weakness. Of course, if they ran into a meteoroid of any size, they'd be ruined—but the chances of that were very slight.

Peal was already outside the hub, dressed in space suit and clinging to a convenient handhold. He came through the lock, carrying his computations,

a small telescope, and an extra spacesuit for Blane. "May need this," he suggested. "Our front end probably won't fit the seal on their hub."

Blane nodded. He should have thought of it. But his chief interest was in the orbit. It had been figured so that they would accelerate away from the station and up from Earth at low thrust, well within the limits of his power. There was a table of times and star angles to locate his correct course. Peal had done an excellent job, far better than Blane had expected.

"I spent two years on the Island," the scientist explained. "I learned a little about astrogation, though I'm no navigator. But this is a simple problem."

Essentially, it was; to make it simpler, it was always possible to make minor corrections, since they had more than enough fuel.

"If the stations were run properly, there'd be a regular service between them," Peal suggested when they were coasting along in their orbit. "It would be cheaper to exchange supplies than to rush up a sudden emergency shipment from Earth. In fact, if a private company had built the first one, there would probably be a dozen stations by now, all connected. And we'd take over the television relay business, too."

At times, Peal sounded like the editorials from a business magazine, but Blane could find no fault with his logic. The fact was that the stations *were* basically service companies, delivering useful services for which they could collect enormous fees without complaints. But they were forced to render most of their service to a military struggle no one wanted and for which no one wanted to be forced to pay.

Peal went on, warming to his theme. "History proves my point, Jerry. The stations have to be too complicated in function and too flexible in purpose to be run properly by men who have to think in terms of Earth politics. Every nation that ever tried controlling a major industrial set-up has found it won't work. They tried socializing railroads, airlines and factories—not to mention farming—and the experiment failed. Every Russian industry today is run independently by its own board who share in the profits, no matter how much theoretical ownership rests with the government. And China is now nothing but a system of state capitalism, whatever they call it there."

"Fine," Blane admitted. "Why didn't private industry build the stations, then?"

Peal grimaced, then grinned. "That's the weak point, of course. You can't sell shares to fund a venture until the public sees the need—and they couldn't see the need of space until military pressure put the stations up and proved they had other values. But now the stations have proved themselves. The government should turn them back to private hands under long-term loans, the same as they turned back factories after the war."

"They won't, though. And it's not just that no power is ever voluntarily given up," Blane pointed out. "They won't sell the stations because they're up here where no government on Earth could tax them. They might eventually, otherwise, but no government is going to lose its profit without getting taxes in return."

For a second, Peal started to argue. Then an expression of surprise crept onto his face. He sat silently through most of the trip. Like most scientists, he'd probably considered himself a fair amateur economist, but he'd overlooked one of the most basic aspects of economy—the fact that governments also had to operate on enough of a profit to pay their executives and bondholders.

At the end of the wide-looping three hour orbit, Blane was surprised and pleased to see that he could locate the Russian station through the telescope. They had made corrections according to Peal's figures, and the scientist had proved to be a better astrogator than could have been expected. Only a tiny corrective blast was needed to bring them into line with the *Sulky*.

As they drew near, Blane stared in amazement. He'd seen pictures, but they had never conveyed the true feeling of the station. Russia had a tradition of building massively for space. Her early ships had been heavy and unsophisticated, relying on strength, size and power. The station was the same. It resembled the *Goddard* superficially, but it was three times as large, and must contain more than twenty times the total volume. It had a solid, substantial look that was indefinable.

The ferry contained a tiny radio, but Blane had not expected it to be useful, since it was adjusted for the frequencies that had been used by the work forces who built the *Goddard*. He reached out and turned it on, expecting nothing. Yet there was a voice coming from it, speaking excellent English. It was a female voice, and a pleasant one.

"Ahoy, space taxi! *Tsiolkovsky* calling taxi. Oh, for Pete's sake, don't you Americans have two-way radio? Wiggle your tail or something so I'll know you're receiving, and I'll give you landing instructions!"

Peal grinned and picked up the microphone. "Ahoy, *Sulky*."

"Ah. So you can answer. Then if you can match our orbit, come beneath the hub. The smallest landing net will fit the nose of your taxi, if our records are correct. You did bring the mercury, didn't you?"

"We brought it," Peal assured her.

"Then in the name of science and humanity, I thank you. And—and I'm so glad to see you, I'll be there to kiss you welcome!"

"There are two of us," Peal started to answer, but she had clicked off. He watched as Blane began jockeying into position, cranking furiously at the little weighted wheel that controlled the angle of the ferry. "Pretty sure we'd come wasn't she?"

"Edwards had a beam antenna on his ship. He could have tipped the *Sulky* off on his way down," Blane said. The little ship was finally lined up and he blasted forward gently against the small landing net. The nose settled firmly into a silicone doughnut that formed a perfect airtight seal. They wouldn't even need to wear spacesuits.

There were three girls and four men waiting for them inside the enormous hub. Six moved forward promptly to begin transferring the cans of mercury, but one girl, shorter, darker and prettier than the others, stepped forward. She kissed both of them—solemnly on both cheeks after the Russian formal fashion. Then she held out her hand.

"I'm Dr. Sonya Vartanian."

Peal introduced Blane and himself. After the handshaking, Blane gestured toward the main station, eager to see it and looking for an excuse. "I'm delighted to know you. But I think I'd better see your commanding officer."

"I'm in command." She said it quite simply. Then at their surprise she chuckled. "We don't have the male chauvinism of America. Besides, all the military officers were below when—when everything was destroyed. But perhaps you'd like to see our station?"

There was a great deal that was crude, and some that seemed to be handmade where American products were smoothly machine made. But generally, it was something to arouse envy in Blane. Obviously, there had been no effort made to save on costs here, and the great Russian boosters had lifted fantastic weights where American engineers had been limited to what ships of lesser thrust would carry. With no restrictions on cost or size, the Russian engineers had simply designed for what they felt desirable, rather than what was possible. The command suite was even equipped with a bar that contained a private refrigerator, though that was now off, due to the need to save power.

The quarters of the staff were spacious, and many showed signs of never having been occupied. The laboratories were beautifully equipped, and again less than a third had ever been used.

"We had great plans—but now we are limited. The threat of war makes even our leaders hesitate to begin so many long-range plans," she explained.

Peal nodded. "You see, Jerry? It's the same here. Waste and inefficiency. This place could make ten times the profit of any other comparable investment, but it's wasted under government control."

Sonya darted him a sudden piercing gaze and stopped in her tracks. Then she laughed uncertainly. "You'll forgive me, Dr. Peal. But those words—they were just what I was going to say."

"You?" Blane stared at her doubtfully. "Isn't capitalistic talk deviationist, at least?"

"Not to an American, and sometimes now not at all." She laughed, as if relaxing from some strain. "We study American economics in our schools, just as we learn your language. Sometimes capitalism seems romantic to us—selling stocks, floating loans, such things. But sometimes I think about what could be done if this were all to be a separate nation, free for all time."

They crossed a great empty section of the station, and Blane recognized that they had already been through there twice before. He saw that Sonya was staring at him intently again as he glanced about more carefully. He moved closer to her, his eyes moving from her face to scratches on the floor and back. She shook her head faintly, and he let the question die unasked.

They ended the grand tour in her office. The power was already on, and the refrigerator was humming. There was no ice, but there was cold water for the drinks she offered them. "You might stay for dinner," she suggested.

Peal seemed embarrassed. "You'll need your supplies ..." he began.

"Supplies?" She laughed at that. "Dr. Peal, here we have supplies to last twice our number for a year, even without a ship. You will stay?"

Blane shook his head. They'd spent too much time already. She accepted the refusal and accompanied them to the waiting taxi, holding out her hand in farewell.

"Sometime, when you need help, remember we are here," she told them. "If there should be any danger or trouble, we are anxious to offer you what we can give."

It was delivered in an almost formal tone, as if now she were rephrasing from her own language.

The trip back was simpler than the first trip, since the ferry now carried no cargo and only half as much fuel. It responded more readily. Peal was silent until they were well away from the *Sulky*. Then he shook his head as if coming out of a brown study.

"Jerry, where do they keep their bombs? We covered every single inch of that station—we went into every room and cranny. I watched to make sure

she wasn't just doubling back. She did, sometimes, but she showed us the whole thing, all the same. And there were no bombs or missiles big enough to dump warheads on Earth. There was one place where they should have been, with what could have been outside release chutes. But it was empty, though there were scratches on the floor where missiles might have stood."

Blane nodded, remembering the place they'd been led across three times. "I know, I saw it. They don't have bombs. They had them, but they're gone. And Sonya Vartanian meant us to see it, too. She didn't quit leading us across the place until she knew I'd guessed."

"Why let us know? So we could report that they've been pulling a colossal bluff at those disarmament meetings? That doesn't make sense."

"No." Blane had been doing his own thinking. "Nobody would believe us—it's incredible, and they'd be sure we'd been duped neatly. They wouldn't dare believe us. And it isn't because Russia is too civilized to use bombs, either; that station was better designed for war than ours, and policies don't change that fast. My guess is that they've been gone from the station two years now."

Peal considered it. "That would be when we spotted the first mass of all their ships together—probably carrying the missiles back to Earth in emergency action. Then that flight that blew up must have been set to carry new missiles up, right?"

Blane nodded. It wasn't a happy idea. It would have taken some very good reason for Russia to remove her missiles during a period of rising tension and hold off for two years before further pressures forced her to resume the idea of stockpiling weapons in space.

He studied the distant *Goddard* through his telescope as they began to draw near. "Maybe I'm wrong, Austin. But they first put warheads out in space a couple of years before we could. And maybe those warheads began to go

through a rapid increase in radioactivity a couple of years before Manners noticed that ours were doing the same. If so, it must have been a pretty serious warning to make the officials disarm the station secretly."

"The girl wanted us to see that the bombs were gone, and she couldn't talk about it. Then she put too much emphasis on that business of offering help if we were in danger." Peel grimaced. "It all adds up."

"How much longer will we have?" Blane asked.

The scientist shook his head. "I don't know, Jerry, and I'm not good enough a physicist to find out."

V

The return was a letdown, after the tension they had been building between them. Blane put the ferry away, leaving no traces of the trip in it, and slipped quietly back into the hub. Things looked miserable now, cramped and forced together, after the spaciousness and richness of equipment on the other station. But he forced that bitterness from his mind.

A Congressman had stated the official policy years before. "Sure, they got something bigger and stronger. But we got the old American spirit. Didn't our boys conquer the whole British navy with nothing but little wooden sailing ships once?" And hence, of course, it didn't matter how badly matched the stations might be. Nobody bothered to comment that the American fleet had grown strong by freebooting, that both sides were using little wooden ships, and that there was never more than a small fraction of the British navy along the American coast. Facts merely got in the way of good sentiment. The Congressman had been elected three times since then and still fought hard to keep any money from getting into space, though he yelled loud and often for the need of teaching the enemy a good lesson.

Blane went to his little room, to bathe in water that was at least hot and clean, and to change into fresh shorts. He had been gone for nearly nine hours, and fatigue had made him look older, but it wasn't too much different from his looks after a sound sleep. He went into the office, yawning. The secretary glanced up, shoved a new mountain of complaints and thunder-

scripts at him, and went on answering the phone. Apparently, he hadn't been too much missed. It wasn't flattering, but he'd expected it.

Routine held him for hours, while he listened to the news from Earth. The Russians were announcing that they had never asked for help from the American supply ships, that the *Tsiolkovsky* was quite safe, and that under no conditions would any political deals be made under threats and pressure. It was done with a nastiness that lent a ring of sincerity to it.

And somewhere, the rumors seemed to indicate, America had modified her stand, and was now making overtures toward helpfulness, which were brusquely refused. There had been an obvious loss of support from some of the smaller nations in the UN, and that must have hurt.

Peal came in, looking more haggard than Blane. The scientist shook his head wearily. "The count is up in the bomb bay. I've been trying to sound some of the chemists out about ways to test, but I don't think we can do it. We don't know what to do or to look for. But I'm convinced now that something is going on inside those casings. It must be some new isotope being created from the uranium by the action of cosmic radiation. Those energies are high enough to cause transmutation. Whatever isotope it is, it must be a neutron emitter, and it's stirring up the uranium, just as increasing the mass does. The temperature around the casings is rising."

"Still no idea of how much margin we have?"

"Not exactly. But I can get some idea from watching how the temperature rises. Maybe a few days, maybe a couple of months." Peal dropped to a couch, rubbing his eyes. "It's getting too hot in there to work without a protective screen, so we can only make short tests. But Manners and I will take turns."

Headquarters was not greatly impressed by the rise in temperature that had been noted, though the reply was longer in coming this time. It simply suggested he stand by for later orders.

That night, a large meteorite fell in Arkansas. It was metallic, and big enough so that several hundred pounds managed to survive the burning

friction of Earth's atmosphere. A large area saw the bright streak across the sky and traced it to where it fell. There its impact had knocked over trees, destroyed a house and the inhabitants, and killed a cow. There was a large hole in the ground where it had hit, and still a trace of metallic fragments around the cup.

Blane picked up the news accounts almost at once on the radio in his office. He switched the circuits around to connect all the speakers in the station and threw the master switch, giving everyone a chance to hear.

It took almost no time for the first reports to come babbling in hysterically, claiming an atomic missile had been sent down from the *Tsiolkovsky*.

The official signal from Headquarters flashed out at Blane, and he listened. They were declaring a general alert, but it wasn't red and there was still a delay. Once it went red, it would mean putting one of the plans already prepared into operation, demanding that he send his few men down into the bomb bay to set the automatic chutes into operation. Then missiles would rain down on Russian cities and bases.

Peal and Manners came in. Manners would have to carry out the orders. Blane glanced at him, and saw doubt and worry etched across the forehead. Could any man start the holocaust going? Or, believing that the *Sulky* would be throwing bombs, as Manners must still believe, could any man refuse such an order?

Blane shook his head faintly as he met Peal's look. There were no bombs on the *Sulky*. And no bombs must fall from the *Goddard*. But in the long run, would it make any difference. There were more than enough land-based missiles to wipe out both countries. And if Blane saw them on his screens, getting set to wipe out *his* nation, could he refuse to order the bombs here into operation?

He threw the side door of the office open and heard the mad action going on outside as men were beaming down the full power of their radio signals, giving the true nature and path of the meteorite, trying to override the frantic chaos already filling the atmosphere.

Then the light winked out. A voice that was weak and shaken came from all the speakers. "Attention. This is official! The object that fell from space has been determined to be a natural meteorite. No attack has been initiated. There is no cause for alarm...."

Blane cut off his speakers and went back into his cabin, shaking with reaction.

This time, there had been no holocaust. This time the alert had never gone red, and sane minds had somehow prevailed. But how long would sanity hold sway in a world where every unnatural accident was a potential trigger for a rain of bombs, a storm that might destroy most of the life on Earth and would certainly end man's adventure into space. It wouldn't really matter whether the stations managed to get off without retaliatory missiles from Earth; once the ships and supply bases were gone, there would be no possibility of continuing life here. The men who fired the missiles from these floating arsenals would be committing a long and horrible suicide. Yet he might have to order it—might reach a stage where he would even want to order it!

Peal was waiting for him with the report on the temperature of the casings when he came into the office the next day. There had been an increase of nearly two degrees, and it began to look as if the rise were an asymptotic one, that might get out of hand so quickly that there would be little warning.

"It's not much of a secret, either," the scientist stated. "I don't think Manners said anything, and I've kept it as tight as I could. But there are indirect ways of noting things going on, and the temperature gages in the hull show signs already. The men who service the bomb bay aren't all fools, either. They can guess there's trouble when they're sent in for only minutes at a time. So rumors are spreading."

Blane nodded. If the rumors got out of hand, things would go to pot in ways that might make it impossible for them to meet an emergency later. He threw the master switch for general summons again, and began speaking slowly, choosing his words with care. He wasn't going to lie, but he couldn't give them full information. He was already violating security to an extent that could bring full official wrath on him.

He told them that there was evidence that radioactivity was leaking from the warheads, though not in any measure to endanger the station at present. He

said simply that there had been some related increase in temperature noted, and that the situation was being studied and reported to Earth, where fuller analysis was possible. It was all true, so far as it went—and the impression was as false as he could make it.

By the time the station was over Denver, where he could contact headquarters on his tightest beam, most of the rumors had died, and the men were discussing the situation without much excitement.

Surprisingly, headquarters took his report and switched him directly to a human, instead of the tape receiver he usually had to deal with. He gave the basic facts, and reported precisely on the fact that he had been forced to inform the crew of the station.

The voice from below sighed wearily across the thousand miles of space. "Quite right, Blane. Panic would be the worst thing you could have. Forget about the violation—we all have to cut that at times. Now, in regard to your basic situation, I'm going to do the best I can for you. But I wouldn't worry about your boiler trouble yet. It will be at least three days before repairs are really necessary, and before then Devlin will be back with you. He has a full grasp of what must be done. And good luck."

The voice cut off.

Blane sat staring at the wall. Three days—it could only mean that there were three days still to go before the runaway radiation inside the casings built up too high for something to be done—whether to dump the bombs or what, he couldn't guess. But that was shaving it pretty thin.

And how sure could he be that they knew what was going on? They had only his coded figures to go by. Yet he had to trust them. For once, he'd be glad when Devlin was back.

He called Manners and Peal in. "Seal off the bomb bay," he told them. "Just stick up a sign making it off limits and spread the word that nobody's to go in until Devlin gets back here—which will be in a couple of days." He grinned at their protests, and shook his head. "And that means off limits to

you, too. Earth says we're safe until Devlin gets here, and he'll have orders. Until then, we can't do anything, so forget the warheads."

It would be a lot easier for the crew of the station to accept than would the sight of Peal and Manners going in and out in constant efforts to check. And there was nothing that their tests could show, anyhow; nobody here knew enough to interpret what the readings meant.

For a change, a sort of lucky accident helped him. One of the pipes in the circulating system got clogged with something that should never have reached the water and burst. It made a mess of most of one deck, and took a full day's cleaning and repairing. That type of misfortune was something the *Goddard* had long since grown used to, and the sight of great scientists working with cooks and power men was always a relief from the routine. Maybe stations should be built to fail in minor ways. If ever a ship was built to cross the vast gulf to another star, it should be as imperfect as safety permitted.

On the surface, everything was routine by the time Devlin's ship came up the next day. Devlin must have more pull on Earth, Blane decided; something had boosted his stock. The ship had taken off from Cape Canaveral—the same ship that had taken him down—in a tricky but successful maneuver. Edwards, of course, had been called in for the job.

Blane had only a few words with the pilot, but he gathered the ship would be standing by to take Devlin off again at some undecided later time.

General Devlin came into the office with brisk, precise steps, and stood looking at Blane with a perfect picture of a military man regarding an inferior. His short body was as straight as a rod, and his head was at precisely the right posture. But his face looked grey, and a muscle under one eye twitched. He motioned sharply as Blane stood up to relinquish the seat behind the desk.

"At ease. Stay where you are. I've been cramped in a hammock for hours, I prefer to stand. I'm not taking over your command this time, anyhow. I'm merely here to execute one order before I have to report back down there. How's the trouble here?"

He listened to Blane's report, but hardly seemed to hear it. He was apparently fully aware of everything that Blane could tell him. When it was done, he nodded. "I was told to fill you in. I'll make it brief. Dr. Peal's theory that ultra hard radiation has caused the transmutation of some of the uranium to a more dangerous isotope is correct. This effects the same results as raising the mass of each segment of the uranium trigger to critical level eventually. But there is still time to save the station, and the level of radiation will not make it dangerous for the squad to handle the missiles; they will be exposed too short a time. I would appreciate it if you would instruct Captain Manners and his men to assemble in the hub in fifteen minutes. I'll join you there."

It wasn't a lot to work on, Blane decided. But he nodded as Devlin went out, pacing toward the coffee in the rec hall. He put through the orders and shortly moved out to join the eight men and Manners. In the hub were stacked a number of boxes. He counted them, and nodded. There was one for each of the missiles.

"Looks like we dump the missiles," Manners suggested, relief heavy in his voice. "Those must be program tapes for the guidance computers on the missiles."

Devlin's voice sounded sharply behind them, bringing them to attention. If he had heard Manners, he gave no sign of it.

"In those boxes are tapes for the missiles. You are all familiar with their installation and the operation of loading the missiles into the outer chutes. Each of you will take one pile of the tapes and repair to the bomb bay. You are to enter there at precisely nine hundred. The bay is hot, but not dangerous for the length of time required to complete this operation. Captain, how long should the operation of moving all bombs into chutes require?"

"About twenty minutes, sir." There were motorized winches that did the work, and the chutes were one of the few pieces of mechanism on the *Goddard* that had not been made shoddily.

"Very good. Then at nine twenty, I shall expect you to emerge from the bomb bay and seal it again. You will then report to Mr. Blane for further instructions."

The orders could have been given just as easily outside the bomb bay, or to Manners alone, Blane realized. The whole affair was too precise, too much by the book. He frowned as he watched Manners and the men pick up the little boxes and move toward the elevators. They were in no great hurry, since they still had fifteen minutes before they were to enter the bomb bay. Then they were gone. And Devlin shuddered faintly and began wiping his face with a kerchief. Something cold shot up from Blane's throat to the roof of his mouth.

"What's the destination on the tapes?" he asked sharply.

Devlin stared at him or through him. Then the stiff body bent a trifle in a faint bow. "I suspect you've guessed it, Blane. They are all set to take an elliptical orbit that will bring them against the *Tsiolkovsky* in mid-Pacific."

"They can't!" But Blane knew that they could be set for just that—they had to be set for such an orbit. With their bomb stock about to become useless in a matter of a few days, and with too little time to replace them after the realization of what was happening, the military mind could decide that the only hope was to eliminate the danger from the other station. It would mean a stalemate in space, and might possibly still leave Russia doubtful enough about the striking power left on the *Goddard* to intimidate her out of retaliating.

"It would wipe men out of space!" he protested. "You've got to cancel the order."

Again Devlin gave the faint bow. "Unfortunately, I have no authority to cancel that order, Mr. Blane. I cannot do so."

Blane felt his fist move from his hip before he realized what he planned. It was an awkward blow, as all activity in nearly zero gravity must be, but it connected. Devlin was lifted from his weak contact with the floor and his head banged savagely against the roof. He drifted back toward the deck, unconscious. Blane caught himself and dashed for the elevator. There was still time to broadcast the facts to the station and to stop the men from entering the bomb bay. After that, he no longer cared what happened to him.

VI

The meeting Blane had called in the rec hall had been brief. Men and women had stared incredulously at him as he told them the facts—all the facts this time. There hadn't even been a vote, since none was needed. Now they were scurrying about, hastily following the orders he had given. Manners was destroying the tapes, the weather men were collecting the reports of future weather that should have been filed within the next few days, and others were gathering what bits of scientific material and notes they could. Edwards had somehow joined them and was already out in the little ferry, heading for the big lunar ship that was fueled and almost completed.

Devlin sat in the hub still. He was conscious now, but the blood on his head ruined what would otherwise have been a fine military posture. He made his slight bow, smiling bitterly in recognition of his helplessness.

"I'm oddly grateful to you, Blane," he said. "But I don't expect you to believe me. And I find I regret what will happen to you and the men here when this catches up with you. What are your plans for me?"

Blane hadn't thought of that. He watched through the port as the ugly, clumsy lunar ship moved toward the ship, to a distance where the ferry could be used to carry them all out to it.

"You can pilot a ship, I remember. Take Edwards' ship and go back to the Island," he decided at last.

Devlin smiled. "I thought of that, too. But with your permission, I'd rather come with you. I'm curious. And I give you my word I shall not interfere in any way. Your case is hopeless, of course—but so is mine."

Blane shrugged. "Come along, then."

Loading everyone into the lunar ship was a horrible period of chaos. It had never been meant to hold such a cargo of goods and people. But somehow room was found, and Edwards began moving out and away from the *Goddard*—the station that was now empty, except for the warheads that were growing hotter with each hour.

"I've called the Island," he said to Blane. "If the message got to anyone except some lumphead, I think they'll be waiting for us."

Blane nodded, but he found little reaction to any news now. He had made his plans in some split moment between striking Devlin and reaching the office. There was nothing now to add to them. There was only a grim determination and the hope that all spacemen must share it—the determination that somehow, men had to stay out here and find an honest destiny in space.

Three hours later, when their long ellipse brought them within sight of the *Tsiolkovsky*, he saw that there were ships around the station. There were no more than half a dozen now, but he could see others approaching. It was impossible for all to leave at once, but the men there had elected to join him, and they had found enough sympathy among the staff of the Island to gain control long enough to accomplish their decision.

The awkward lunar ship came to a reluctant stop less than half a mile from the station, and Blane began picking those who were to go with him aboard the little ferry that was in tow. Manners and Peal and two others. He was looking for a sixth when Devlin moved into the group. Blane started to order him aside, and then shrugged.

They were almost as crowded in the taxi that Edwards piloted as they had been on the lunar ship. But there was no thought of that. The others were taking their cue from Blane, and Blane was simply waiting, frozen in his determination until events could shape his moves.

The landing net snapped around them, and they settled into the silicone ring, and then began moving into the huge hub of the *Sulky*.

At least a dozen people were waiting there—too many, Blane realized. He hadn't bothered to consider the size of the group he must meet. But he disregarded that.

Sonya Vartanian moved forward to greet him with the double kiss and handshake. Her eyes were unreadable, but her voice was warm. "Welcome,

gentlemen. I am delighted that you remembered our offer of aid in time of trouble. You have our assurance that—"

Blane cut her off with a hasty gesture. He wanted no speeches from her. It had to be done at once, or forgotten, as he had planned it. His mind had no second line of action. He began to speak authoritatively;

"In the name of the free territory of space, I seize this ship and all that is on it," he continued coldly, "I sever all ties any may have with Earth herewith. I ban all military operations from space. I declare that no nation may own property in space, but may only trade according to the just laws and practices that shall be henceforth established for space. I—"

He was wound up to the point where he could not stop, though there was nothing more to say at the moment.

But a sudden sound choked off his words. It was a shout from those who had come to meet the crew from the *Goddard*. It was a long, surprised crescendo that slowly became a cheer.

Sonya leaned forward, grasping his hands. "Thank God," she cried in his ear. "Oh thank God. I was so afraid you wouldn't see it."

He blinked, beginning to feel foolish. "You mean that you agree? Without resistance?"

"We've been trying to find some way to make it happen for two years—ever since our commander refused to permit our decaying missiles to be used against your station," she told him. "But we never really believed it could happen."

Blane knew that he had never believed it, either.

He pulled her closer, beginning to smile again. "In the name of the free territory of space!" he said, and kissed her.

The blaze in the heavens that had signalled the end of the *Goddard* was less than twelve hours old. It had been a magnificent funeral pyre to an epoch, but it had not yet ended the methods of diplomacy. It had merely forced faster action.

The Premier of Russia and the President of the United States sat together, trying to keep their voices down and yet hear each other over the noise and confusion of the assembly hall in the UN building. They were surrounded by guards, as usual, and the television cameras were focused on them. But they had so far been unable either to agree or disagree. They could only wait until the time announced had arrived, as most of the world was now waiting.

Then the great system of amplifiers and speakers went into operation, and quiet began to descend over the hall.

There must have been a greeting of some formal kind, but few heard it. Jerry Blane's tired voice was already setting forth his written statement of demands when the quiet was sufficient for him to be heard. He read with the voice of a man not used to making a written speech sound natural, but nobody noticed.

The announcement of the facts was obvious, but it took on added power from the brevity that compressed everything into a single focus. America had lost a station and Russia had no supply ships. There was a supply base on Johnston Island, but the ships were all in space. Earth was completely cut off from contact with space for months to come.

And Earth could no longer exist without that contact. Her next weather reports were needed within the week, and without them the damage to crops grown dependent on them might result in famine for much of Earth. Certain drugs had to be made in space. There were hundreds of needs, without which the economy of Earth would collapse. Today, in a real sense, Earth could exist only by the use of a station in space.

But the station could exist for a longer time without Earth. There was food and supplies for more than a year. They were prepared to wait, if need be.

"You cannot use force," Blane's voice stated flatly. "For the first time, the governments of Earth cannot fall back on destruction when everything else fails. To destroy us would make your economic collapse inevitable now. You cannot go back to your past or the savage rules of your past. You can only meet us honestly and concede the just demands we propose."

Many were surprised at the proposals—the joint work of two years of thought on the *Tsiolkovsky* and a final flash of insight on the part of Blane. They wanted recognition from the UN that they were an independent territory. They wanted to incorporate as an independent stock company on Earth, under direct UN charter. For that, they were willing to pay reasonable taxes on operations done within any country. They were willing to pay a reasonable price, to be settled by a committee of neutral nations, for the two stations, for the ships—and even for the Russian ships that were destroyed—and for complete sovereignty over Johnston Island, which would now be worthless to Earth. They would pay for this by the issuing of stock, which could be redeemed in time through the profits that were easily provable as more than adequate to meet their debts. And they were to have full control of further ventures and services to be transacted on the station. Weather predictions would be on a subscription basis, research on the station would be by lease, and other services could be adjusted to a fair market value.

There was more, but much of it was only repetition to make sure all was understood. It finished with a simple request for a quick decision, since no more business could be done with Earth until the agreements had been reached.

The President nodded. "You'll agree?" he asked.

"What else can we do?" the Premier asked in return. "He's right. We can't continue today without the services we're used to from space. A series of accidents has left us no choice."

The President settled back, apparently satisfied. But he was less sure. Had there been accidents involved? Some man must have hated war in space enough to sabotage a fleet of ships. Other men had hated that same war enough to break all discipline and strike out against a whole planet. And men and women on two separate stations had so detested the thought of being crushed in a surface struggle that they had independently schemed for this proposal.

He let his eyes rest on the delegate from Israel who was yielding to the delegate from Saudi Arabia. It didn't matter who made the resolution to accept the proposal of Blane on a tentative basis. There would be no veto possible now.

And on Earth, the tension was relaxing already. Perhaps now, even the surface enmities could be settled in time.

The Fifteenth Space Disarmament Conference was ending.

THE END

*** END OF THE PROJECT GUTENBERG EBOOK THUNDER IN
SPACE ***

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